

Energy Yield Prediction of Bifacial Perovskite/Silicon Tandem Photovoltaic Modules

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Bifacial perovskite (PVK)/crystalline silicon (c-Si) tandem photovoltaic (PV) modules provide an effective strategy to further improve the efficiency and energy yield (EY) of c-Si PV modules. In this work, the energy outputs of bifacial tandem PV modules under outdoor conditions are analyzed by combining optical modeling, one-diode equivalent circuit model, module tilt angle, and meteorological data, and a detailed comparative study with bifacial silicon heterojunction (SHJ) PV modules is also performed. The bifacial PVK/c-Si tandem modules exhibit higher EY in locations with strong direct normal irradiance regardless of the ground type with any albedo, while bifacial SHJ modules exhibit similar or even higher EY than tandem modules in locations with strong diffuse horizontal irradiance (DHI) such as Chengdu. Considering factors such as EY and leveled cost of electricity, E_{g_PVK} in the range of 1.58–1.62 eV is suitable for most application scenarios, while the deployment of bifacial SHJ PV modules is preferred in areas dominated by DHI, as represented by Chengdu. Herein, important implications for policy making in determining the cost-effective type of PV modules are provided to generate renewable electricity.

passivating contact has reached 26.81%,^[3] which is practically approaching the limit of 29.4% (with the Auger recombination taken into account).^[4] This reality has stimulated an explosive research interest in both academia and industry to develop tandem PV in which c-Si bottom cell is stacked with wide-bandgap top cells to better utilize the full solar spectrum and thus to further boost PCE.

Given the tunable bandgap, high band-edge absorption, and cost-effective fabrication process,^[5,6] perovskites (PVK) solar cells are considered as the most promising top cell candidates for silicon-based tandem solar cells. Following the pioneering report in 2015,^[7] PVK/c-Si tandem solar cells have experienced rapid progress and have recently reached an efficiency of 33.2%,^[8] which has far exceeded the theoretical limit of single-junction c-Si solar cells. Recently, remarkable progress has been made to improve the stability of PVK materials with

1. Introduction

Photovoltaic (PV) power generation is an important driving force in solving the global energy crisis and achieving global carbon neutrality. Crystalline silicon (c-Si) solar cell technology has been developed over the years and has dominated more than 95% of the PV market.^[1,2] The power conversion efficiency (PCE) of silicon heterojunction (SHJ) cells deploying the concept of

effective strategies proposed to ensure the long-term stable operation of PVK-based solar cells.^[9–14]

It should be noted that the PCE of solar cell obtained in the laboratory measured under AM1.5G standard spectrum cannot represent the energy yield (EY) in real-world conditions. Instead, for cells and modules operated under outdoor conditions, it is obliged to consider the direct normal irradiance (DNI), diffuse horizontal irradiance (DHI), and rear-side albedo

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radiation to correctly estimate EYs, which together with the leveled cost of electricity (LCOE) are the key indicator of the outdoor performance or cost advantage of different PV modules.

Hörantner and Snaith developed a rigorous optical model and combined it with one-diode equivalent circuit model to obtain the PCE of planar monolithic PVK/*c*-Si tandem solar cells and estimate the EY for different environments and axis tracking designs.^[15] They concluded that planar monolithic PVK/*c*-Si tandem solar cells could produce more EY than single-junction SHJ solar cells by 30%. This pioneering work attracted widespread attention, but did not take into account the optical losses of the modules, and the cells were modeled in a planar structure which is inconsistent with the textured *c*-Si commonly used in PV industry. Following work then introduced textured *c*-Si to calculate the EY of PVK/*c*-Si tandem solar cells and discovered that the textured *c*-Si surface could efficiently improve the EY of tandem PV modules.^[16–19] After that, the climate-dependent EY of tandem modules on a global scale was investigated, which demonstrated that the monofacial tandem modules are more advantageous than SHJ modules in terms of EY.^[20] Assuming comparable lifetime of PVK materials with that of *c*-Si cells, the LCOE of high-efficiency PVK/*c*-Si tandem modules is predicted more advantageous than that of single-junction *c*-Si modules.^[21]

In the previous work, the EY and related characteristics were mainly conducted for monofacial PV modules in which SHJ was typically used as the bottom cell.^[15–17,20,22,23] However, SHJ solar cells naturally possess bifacial light-receiving feature that can effectively absorb diffuse light reflected by the ground (called albedo) and by the atmosphere and surroundings. This bifaciality will result in an increase in the EY of about 25–30%.^[24–26] According to the International Technology Roadmap for Photovoltaic, the market share of bifacial modules is expected to increase from 50% in 2022 to about 85% by 2032.^[27] Without doubt, bifacial tandem solar cells will become a competitive player in the PV market in the future. Therefore, it is of great significance to study the EY of bifacial PVK/*c*-Si tandem modules for large-scale deployment. Some of the recent works have predicted the EY of bifacial tandem solar cells,^[19,28–30] in which the focus is mainly on how the different albedo affects the optimal bandgap and thickness of PVK layers. Little attention has been paid to performing EY and LCOE analysis for bifacial tandem PV modules installed in more typical regions and their comparison with bifacial SHJ PV modules are barely studied.

In this work, we analyzed the EY characteristics of bifacial tandem modules in different regions (with typical solar irradiation) and under different albedo conditions with respect to that of bifacial SHJ modules. In most regions, the EY of bifacial tandem modules outperforms the bifacial SHJ modules. And, the optimal E_{g_PVK} is suggested to be 1.62 eV or even lower to produce the highest EY for bifacial PVK/*c*-Si tandem modules with the presence of albedo. Compared to wide-bandgap PVK ($E_{g_PVK} = 1.67$ eV or larger), narrow-bandgap PVK materials are featured with lower voltage loss, higher stability, and maturer fabrication process.^[31–34] In particular, it is worth noting that in Chengdu area, which is dominant with DHI, bifacial tandem modules exhibit similar (albedo $\leq 35\%$) or lower (albedo $> 35\%$) EY than that of bifacial SHJ modules. Given the considerations of

EY and LCOE, tandem modules are suitable for most areas except where DHI dominates.

2. Experimental Section

2.1. Optical Simulation

The complete structures of the three types of solar cells, that is, bifacial SHJ (as reference), bifacial, and monofacial PVK/*c*-Si tandem, are shown in Figure S1, Supporting Information. The monofacial PVK/*c*-Si tandem was constructed by placing a 500 nm thick Ag on the rear side of the *c*-Si bottom cell to block the incident light from the rear side, which at the same time helped to recycle the incident near-infrared (NIR) photons from the front side. To evaluate the power generation and annual EY under outdoor conditions, the optical characteristics of the corresponding modules as illustrated in Figure 1a are first simulated. The optical simulations were carried out using SunSolve,^[35] which combined ray tracing with a coherent transfer matrix-based thin-film simulator and reproduced the optical absorption of each layer based on the optical parameters of each layer. For the optical parameters of PVK, we referred to the previously reported complex refractive indices at 1.557 eV (MAPbI₃), 1.62 eV (Cs_{0.17}FA_{0.83}Pb(Br_{0.17}I_{0.83})₃), and 1.67 eV (Cs_{0.25}FA_{0.75}Pb(Br_{0.2}I_{0.8})₃) by Manzoor et al.^[36] and simulated the other bandgaps (i.e., 1.58, 1.60, and 1.70 eV) by assuming a perfect horizontal shift along the energy axis (Figure S2, Supporting Information), which was confirmed in the work of Manzoor et al.^[36] The detailed method is presented in the Note S1, Supporting Information. The optical parameters of the remaining layers were obtained through literatures.^[36–47]

2.2. Electrical Model

Assuming that the absorption of one incident photon generated one electron–hole pair, and all electron–hole pairs contributed to the photocurrent,^[48–50] the short-circuit current densities (J_{sc} s) of the PVK (J_{sc_PVK}) and *c*-Si (J_{sc_c-Si}) subcells could be obtained by $J_{sc} = e \times \int \frac{1}{hc} \text{Abs}(\lambda) I_{AM1.5G}(\lambda) d(\lambda)$, where e is the elemental charge, h is the Planck's constant, c is the light speed in vacuum, $\text{Abs}(\lambda)$ is the absorption of each material, and $I_{AM1.5G}(\lambda)$ is the AM1.5G solar spectral irradiance (ASTM G-173-03).^[51] Note that the values of wavelengths-dependent $\text{Abs}(\lambda)$ of PVK and *c*-Si were equivalent to the external quantum efficiencies (EQEs) of PVK and *c*-Si subcells based on the aforementioned assumption.

The one-diode equivalent circuit model was employed to model the current density–voltage (J – V) characteristics of PVK/*c*-Si tandem solar cells/modules with the help of LTspice software.^[52] To obtain the specific J – V curves, the photogenerated current (J_G), dark saturation current (J_0), ideal factor (n), series resistance (R_s), and shunt resistance (R_{sh}) were used as input parameters, where the J_G was obtained from the J_{sc} of the cell or module under AM1.5G illumination. The diode parameters of *c*-Si bottom cell (J_0 , n , R_s , R_{sh}) were extracted from the J – V curves in the reference.^[53] In the case of PVK top cell, the J_0 for each bandgap was calculated based on the blackbody radiation at 300 K and the corresponding EQE.^[54] The calculation details are described in the Note S2, Supporting Information.

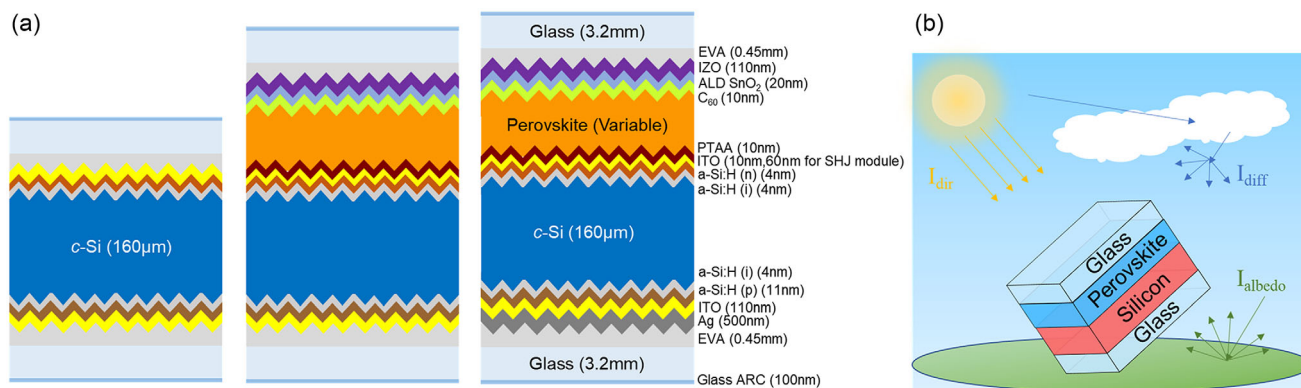


Figure 1. The schematic diagrams of a) the bifacial silicon heterojunction (SHJ) module, bifacial and monofacial perovskite (PVK)/c-Si tandem modules, b) the direct (I_{dir}), diffuse (I_{diff}), and albedo (I_{albedo}) irradiance sunlight irradiating on the bifacial PVK/c-Si tandem module.

The remaining parameters (n , R_s , R_{sh}) were extracted from the J - V curve of a single-junction PVK with an efficiency of 21.4% reported by Fan et al.^[55] Assuming that all PVK solar cells with different bandgaps had the same n , R_{sh} , and R_s , the diode parameters used for the SHJ solar cell and PVK solar cell simulations are listed in Table S1, Supporting Information. The PCE could then be rationally obtained from the maximum power point (MPP) divided by the incident power.

2.3. Calculation of Annual EY

The annual EY was evaluated based on PV modules at different locations with **fixed tilt angles**. The direct and diffuse sky irradiance at different locations were derived from the hourly resolved third edition of typical meteorological year data, in combination with the cloud model and the simple model of atmospheric radiative transfer of sunshine.^[56,57] The front-side illuminations on cells and modules were determined by the geographical location, season, and weather, and **we assumed that the spectra varied equiproportionally with the light intensity**. Since c-Si cells could efficiently absorb short-wavelength light below 900 nm, the reflection from the ground could hardly contribute to the photocurrent of PVK top cell,^[26,28,58,59] and thus the variation of the **reflected spectrum had practically little effect on the EY of the monolithic bifacial tandem PV module.**^[19] The rear-side illumination on the bifacial PV module was determined by a combination of direct solar irradiance, diffuse sunlight scattered in the surrounding and atmosphere as well as ground albedo, **assuming that all radiation received by the rear side was isotropic.**

3. Results and Discussion

There are three main sources of sunlight that irradiate on PV modules,^[60] that is, direct light from the sun (I_{dir}), diffuse sunlight scattered around and in the atmosphere (I_{diff}), and diffuse sunlight reflected from the ground (I_{albedo}), as schematically shown in Figure 1b. In general, the I_{dir} and I_{albedo} contribute to the front- and rear-side irradiance on the PV modules, respectively, while the I_{diff} can simultaneously interact with the front and rear side of the modules.

The optical simulations are carried out on SHJ, bifacial, and monofacial tandem solar cells, the structures of which are illustrated in Figure S1, Supporting Information. According to previous reports, the optimal bandgap of PVK (E_{g_PVK}) top cell in the monofacial tandem device ranges from 1.65 to 1.70 eV under standard test condition (AM1.5G).^[15,18,22] As such, an E_{g_PVK} of 1.67 eV is first employed for the top PVK absorber. Assuming the absorption of one incident photon generates one electron-hole pair, and all electron-hole pairs contribute to the photocurrent, the J_{sc_PVK} and J_{sc_c-Si} are obtained by calculating the absorption in 300–1200 nm with an interval of 10 nm as mentioned in the method section. The optimized PVK thickness is found to be 820 nm in the **monofacial** tandem solar cell, under which the J_{sc} mismatch between PVK and c-Si is 0.01 mA cm⁻² (Figure S3, Supporting Information). **In the subsequent modeling, the thickness of the PVK layer is fixed to 820 nm unless otherwise specified.**

The absorption spectra of each functional layer composing the monofacial and bifacial tandem solar cells with a PVK bandgap of 1.67 eV are represented in Figure S4a,b, Supporting Information, where only front-side illumination is considered. Due to the absence of Ag back reflector in the bifacial tandem solar cell, the reuse of light in the NIR region is significantly suppressed, resulting in a reduced absorption in both c-Si and rear ITO. The J_{sc_c-Si} is reduced from 19.90 to 19.24 mA cm⁻² (also see Table S2, Supporting Information). The indirect bandgap characteristic of c-Si gives rise to a weak band-edge absorption and thus reduced J_{sc_c-Si} .^[61] Similar result is also observed in PV modules (see Figure 2a,b) where the equivalent J_{sc_c-Si} is reduced from 19.17 to 18.42 mA cm⁻² with a marginal change of J_{sc_PVK} when only front-side illumination is considered. The absolute current loss values in encapsulated modules (Figure 2a,b) compared with unencapsulated solar cells (see Figure S4a,b, Supporting Information) mainly originate from the increased front-side reflection and parasitic absorption from cover glass and ethylene vinyl acetate copolymer (see Table S2, Supporting Information for more detailed comparison). The same trend of cell-to-module loss is also found in bifacial SHJ solar cells and modules as shown in Figure S4c,d and Table S3, Supporting Information.

Due to the reduced absorption of the c-Si in the bifacial tandem PV module, the J_{sc_c-Si} becomes the limiting factor, and

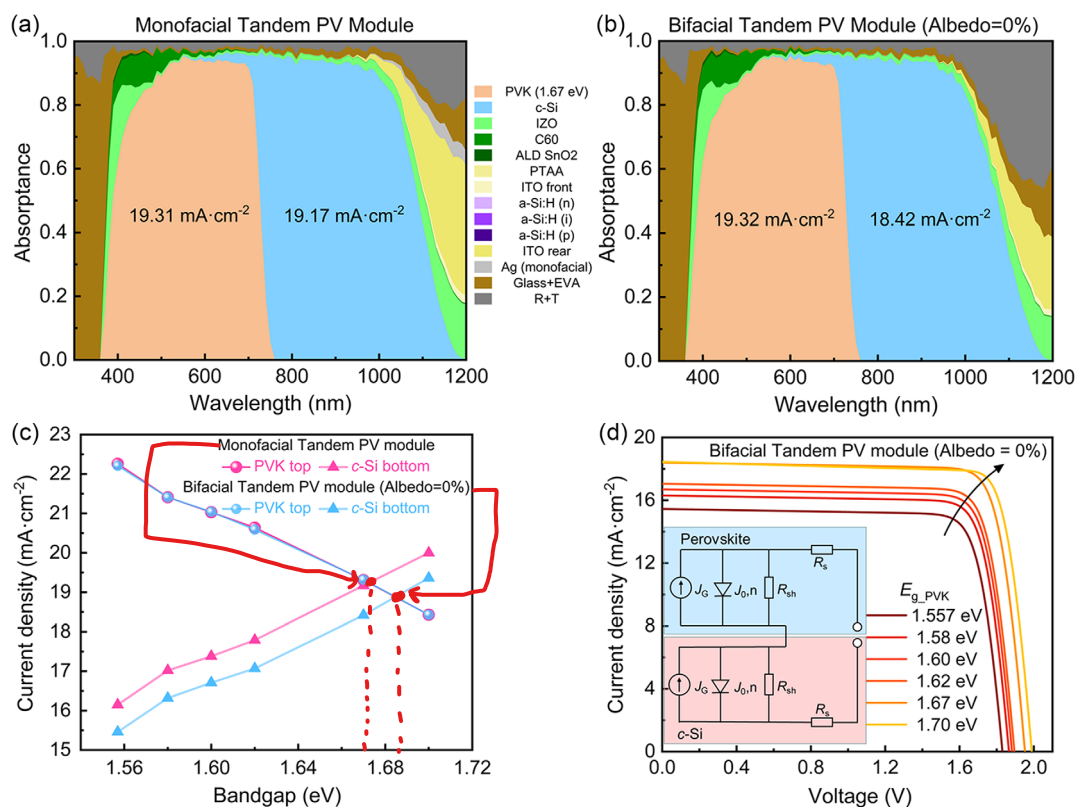


Figure 2. The absorption of each layer as well as the reflection and transmission spectra of a) monofacial and b) bifacial tandem modules. c) The $J_{sc,PVK}$ and $J_{sc,c-Si}$ as a function of $E_{g,PVK}$ in monofacial and bifacial tandem modules. d) The plot of current density–voltage (J – V) curves of bifacial tandem modules with different $E_{g,PVK}$ under AM1.5G (albedo = 0%), with the arrow pointing to increasing $E_{g,PVK}$. The inset shows the equivalent circuit of the tandem solar cell.

therefore a higher $E_{g,PVK}$ is required to reduce the current mismatch. The $J_{sc,PVK}$ and $J_{sc,c-Si}$ are then calculated as a function of $E_{g,PVK}$ from 1.557 to 1.70 eV on both monofacial and bifacial tandem PV modules under AM1.5G front-side illumination as shown in Figure 2c. One can clearly see that the optimal PVK bandgap is increased to ≈ 1.69 eV for the bifacial case. The variation of $E_{g,PVK}$ will affect both the subcell J_{sc} and open-circuit voltage (V_{oc}), and thus the PCE of the tandem PV modules (Figure 2d). The equivalent circuit of a tandem solar cell is illustrated in the inset of Figure 2d, where the one-diode models are used for each subcell. For the c-Si bottom cell, the diode parameters (see Table S1, Supporting Information) are extracted from the J – V curve of an SHJ solar cell with a PCE of 25.11%,^[53] which fits our modeled curve (Figure S5, Supporting Information) within the acceptable error range. One notes that this efficiency is a reasonable value since it is achievable in mass production, considering the recent record efficiency of 26.5%.^[62] The J – V curves of PVK solar cells (Figure S6, Supporting Information) with different bandgaps are obtained by assuming the same n , R_{sh} , and R_s diode parameters (see Experimental Section and Table S1, Supporting Information).^[15] Figure 2d shows the J – V curves of the series-connected bifacial tandem modules with different $E_{g,PVK}$. Figure 2c depicts that the bifacial tandem solar cells having PVK subcells with 1.67 and 1.70 eV bandgaps render almost the same J_{sc} , where the current limiting subcell is

switched from c-Si to PVK. But thanks to the higher V_{oc} , a higher module efficiency of 30.45% is achieved at an $E_{g,PVK}$ of 1.70 eV. Thus, the current matching point should be for $E_{g,PVK}$ of around 1.69 eV, which can deliver even higher module efficiency.

In the previous discussion, however, the effect of rear-side irradiation on bifacial tandem modules was not considered, that is, albedo = 0%. The bifacial concept is known to be widely employed in c-Si modules. The SHJ solar cell with a bifaciality of more than 90% can produce a significant extra energy gain thanks to the rear-side power generation, which has been known to vary as a function of albedo.^[63,64] As such, in the following discussion, the rear-side illumination will be incorporated to assess the potential of bifacial tandem modules in generating additional power. The front illumination is kept at AM1.5G and the isotropic irradiance of the rear side is determined by the different albedo ($I_{back} = \text{albedo} \times I_{AM1.5G}$).

A term so-called “pseudo-EQE” is employed to evaluate the photocurrent of the bifacial module, which is defined as the flux of electrons excited by both front- and rear-side illumination divided by the flux of photons incident on the solar cell from the front side. Take the $E_{g,PVK}$ of 1.60 eV as an example, Figure 3a shows the pseudo-EQE curves for bifacial tandem PV modules at albedo = 0% and 100%. As the c-Si bottom cell receives additional reflected light at albedo = 100%, its pseudo-EQE values exhibit a sharp increase over the entire

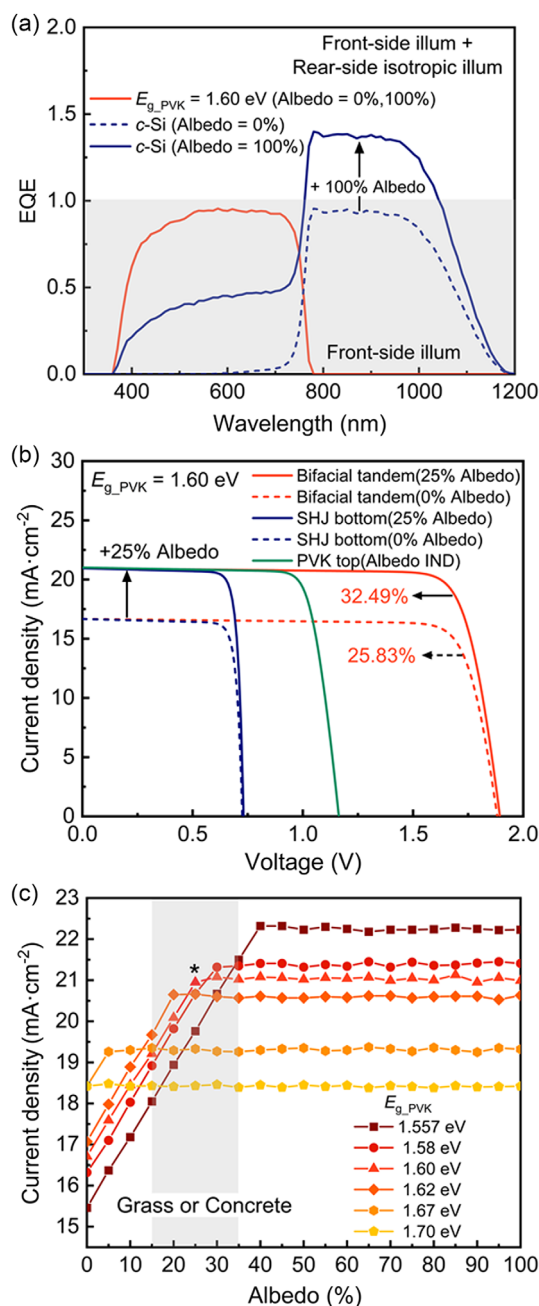


Figure 3. a) Pseudo external quantum efficiencies (pseudo-EQE) curves for bifacial tandem modules at albedo = 0% and 100%. b) J - V curves of bifacial tandem solar module, together with corresponding PVK top cell ($E_{g_PVK} = 1.60$ eV) and c -Si bottom cell for albedo of 0% and 25%. c) The J_{sc} of bifacial tandem modules as a function of albedo and E_{g_PVK} . The region highlighted in gray represents the albedo of typical ground condition such as grass (albedo = 25%) and concrete (albedo = 35%). The point marked with star represents the current matching point at an albedo of 25% and E_{g_PVK} of 1.60 eV. "illum" and "IND" are abbreviations for illumination and independent, respectively.

spectral range and are greater than 100% from 770 to 1040 nm. The pseudo-EQE plots for other albedo values are also shown in Figure S7, Supporting Information, for references. Figure 3b

represents the variation of the J - V curves for the bifacial tandem module with $E_{g_PVK} = 1.60$ eV. The bifacial tandem PV module shifts from a large current mismatch ($J_{sc_mismatch} = 4.33 \text{ mA cm}^{-2}$) at albedo = 0% to negligible current mismatch ($J_{sc_mismatch} = 0.03 \text{ mA cm}^{-2}$) at albedo = 25%, and the PCE increases from 25.83% to 32.49%.

As learned from the effect of albedo and E_{g_PVK} on J_{sc} shown in Figure 3c, it is found that the smaller E_{g_PVK} , the more obvious the increase of albedo in improving the J_{sc} of the tandem module. In other words, for bifacial tandem modules with large albedo, PVK subcell with a lower bandgap is preferred, which is consistent with previous reports.^[29,30,58] The region highlighted in grey corresponds to the albedo for typical ground conditions, such as green grass (albedo = 25%) and concrete (albedo = 35%).^[65,66] The PCE of the tandem module of all E_{g_PVK} efficiently improves after implementing green grass (albedo = 25%) at the rear side (Figure S8, Supporting Information) and the current matching is achieved for $E_{g_PVK} = 1.60$ eV giving rise to maximum PCE of 32.49%.

It is worth noting that with an E_{g_PVK} of 1.70 eV, PVK is the current-limiting subcell and increasing the albedo does not change the J_{sc} of the tandem module. Instead, the increase in current mismatch causes an increase in fill factor (FF), which increases the PCE of the module from 30.45% to 30.94% (Figure S8f, Supporting Information). This is consistent with the previous report which states that the lowest value of FF is located at the current matching point.^[20,29] For bifacial tandem modules with other E_{g_PVK} s, one also sees an increasing FF with rising current mismatch after increasing albedo (see Figure S9, Supporting Information).

The previous discussions have been conducted under ideal conditions without considering the location as well as the corresponding solar irradiance. In the realistic outdoor operation, a variety of factors including site conditions (i.e., albedo, average sun-hour, elevation from sea level, latitude, DNI, DHI, etc.) and PV module specifications (i.e., bifaciality, efficiency under standard test condition, E_{g_PVK} , etc.) can strongly influence the power generation and EY of the bifacial tandem PV modules.^[15,67–69] It is thus a challenging task to match J_{sc_PVK} and J_{sc_c-Si} in different geographical locations and under seasonal variations. To evaluate the performance of bifacial tandem modules in different regions, the calculation of EY is conducted at six sites with typical insolation conditions, that is, Chengdu (30.7°N, 104.1°E) (subtropical humid monsoon climate), Shanghai (31.4°N, 121.5°E) (northern subtropical monsoon climate), London (51.5°N, 0.1°W) (temperate maritime climate), Yinchuan (38.5°N, 106.2°E) (temperate continental climate), Lhasa (29.7°N, 91.1°E) (highland temperate semiarid monsoon climate), and Los Angeles (34.1°N, 118.2°W) (subtropical Mediterranean climate) (see Figure S10, Supporting Information, for meteorological data).

Considering the cost of practical installation, the modules are designed with a fixed tilt angle. By summing the hourly EY (that is, the product of maximum power (P_{MPP}) and 1 h) over the entire year, one obtains the annual EY ($= \sum P_{MPP} \cdot 1 \text{ h}$). The shadow under the PV module that reduces ground reflection could be neglected at a module installation height of 1 m or more.^[19] Since the photocurrent of PVK subcell is hardly affected by the reflected light of <900 nm as mentioned earlier, the variation in the spectral distribution of reflected light has little effect on the

PVK subcell and thus the use of average albedo is justified for outdoor EY assessment.^[28] To acquire a more realistic EY, temperature-dependent module performance should be considered, which needs to account for ambient temperature, wind speed, as well as heat absorption and conduction between layers of PV module in different geographic environments. In addition, the scattered irradiation received from the rear side also affects the module temperature, making it more difficult to model the module temperature. **To simplify the model, the shading effect of the surrounding modules, wind speed, temperature, and module degradation as well as their temperature coefficients are not considered in the simulation.**

The EYs of bifacial tandem PV modules in different regions with respect to the tilt angle and E_{g_PVK} are represented in **Figure 4**, where the albedo is set to a constant value of 25% (typical ground condition). The module with the highest EY of 622.0 kWh m^{-2} is found in Los Angeles, where the optimal E_{g_PVK} and tilt angle are around 1.62 eV and 32° , respectively. The highest EY in Chengdu, on the contrary, is only 242.5 kWh m^{-2} . Both Lhasa and Yinchuan are characterized by good solar irradiance,^[70,71] but since Lhasa shows higher DNI and DHI than Yinchuan (see Figure S10, Supporting Information), the bifacial tandem module yields a higher EY in Lhasa. Despite a better global horizontal irradiance (GHI) in Lhasa than that of Los Angeles, Lhasa yields a lower EY due to that its share of DHI in summer (mostly from June to September) being larger than that in Los Angeles (see Figure S10 and S11, Supporting Information). The optimal tilt angle of the modules is generally slightly lower than the latitude and positively correlated with latitude, that is, the higher the latitude, the higher the tilt angle. By

comparing the previous results, we found that the DNI can significantly affect the optimal E_{g_PVK} . In regions with high DNI such as Los Angeles, Lhasa, and Yinchuan, the E_{g_PVK} between 1.60 and 1.64 eV can match well with *c*-Si bottom cell to obtain a high output power and thus EY. In regions with high DHI (e.g., Shanghai, London, and Chengdu), the reflection from the front surface of the module is increased due to the oblique incidence, resulting in less photons reaching the PVK absorber. In addition, the DHI can also contribute to the photocurrent of *c*-Si cell, further enlarging the current mismatch between top and bottom subcells. In this case, lower E_{g_PVK} ($1.56\text{--}1.60 \text{ eV}$) is better employed to achieve the current matching by enhancing the PVK absorption of photons with longer wavelengths to compensate for the reflection loss from the front side and the photocurrent gain of the *c*-Si bottom cell.

Finally, we modeled the EY of the bifacial PVK/*c*-Si tandem modules by varying the albedo to simulate different ground conditions (albedo = 9% for gray sandstone, albedo = 25% for grass, albedo = 35% for concrete, albedo = 64% for bright sandstone, and albedo = 88% for snow^[72]) for the above locations with different meteorological conditions (see **Figure 5** and S12, Supporting Information). The EY of the bifacial SHJ module is also calculated as a reference (see the gray bars in Figure 5a–c and S12, Supporting Information). For bifacial tandem modules with all values of E_{g_PVK} , increasing albedo leads to improved EY: the increase is more significant for the tandem modules with lower E_{g_PVK} . For tandem modules with E_{g_PVK} of 1.70 eV , despite the increasing current mismatch with increasing albedo, the EY still shows slow enhancement with higher albedo. This is mainly due to the increase in *FF* caused by the increase in

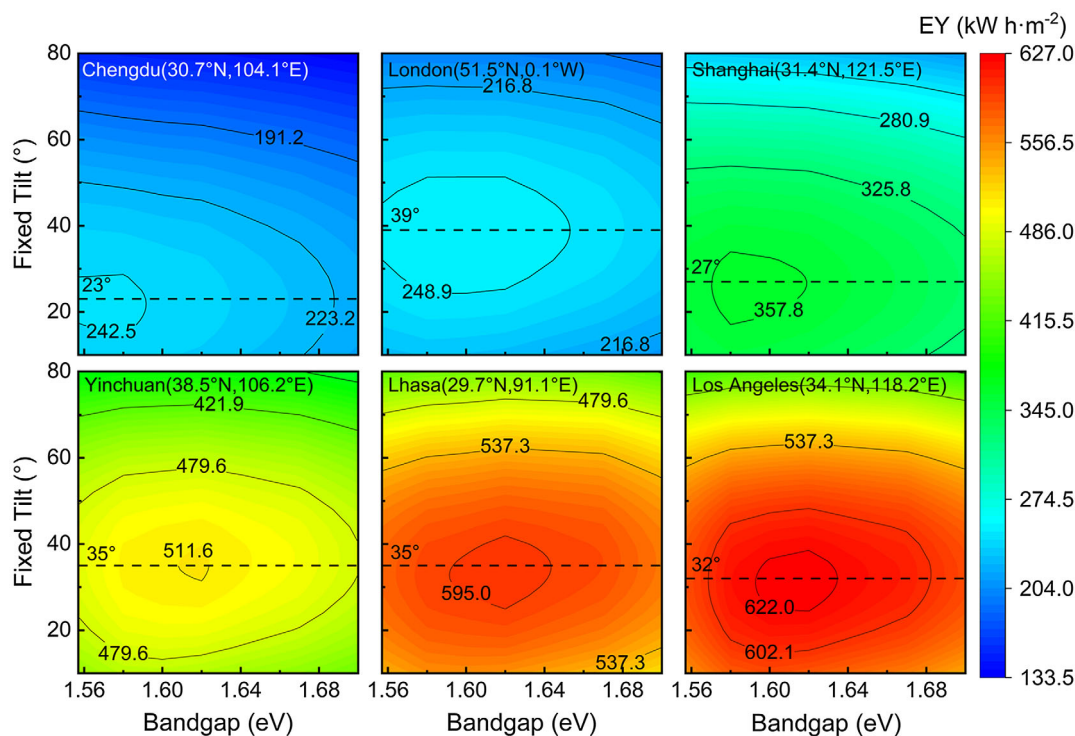


Figure 4. The energy yields (EYs) of bifacial tandem modules in different regions (Chengdu, Shanghai, London, Yinchuan, Lhasa, and Los Angeles) with respect to the tilt angle and E_{g_PVK} .

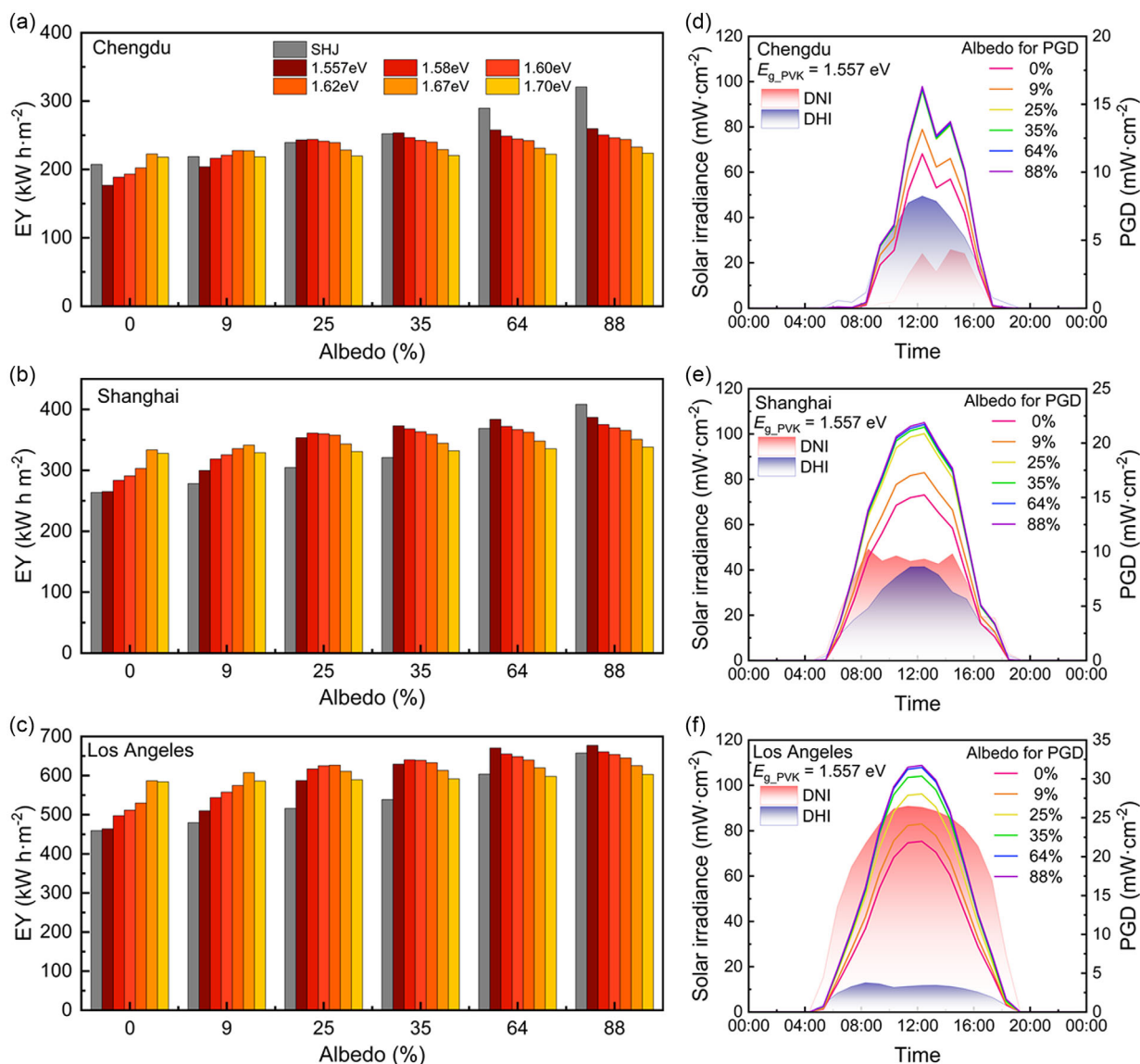


Figure 5. A comparison of EY for bifacial SHJ modules and bifacial PVK/*c*-Si tandem modules in a) Chengdu, b) Shanghai, and c) Los Angeles. The power-generation density (PGD) of bifacial tandem modules ($E_{g_PVK} = 1.557$ eV) at different albedo conditions in a summer day in d) Chengdu, e) Shanghai, and f) Los Angeles, along with their meteorological characteristics.

current mismatch.^[29] It is noted that for albedo = 0%, although the highest *PCE* is achieved with an E_{g_PVK} of 1.70 eV under standard AM1.5G condition (Figure 2d), the highest EYs is achieved by the tandem modules with an E_{g_PVK} of 1.67 eV. This is because, during outdoor operation with albedo of 0%, the diffuse light from the surrounding atmosphere not only shines onto the PVK top cell but also contributes to the photocurrent of the *c*-Si bottom cell from the rear side. In this case, the tandem cells with an E_{g_PVK} of 1.67 eV feature better current matching and therefore higher EYs. In all locations we studied, E_{g_PVK} of 1.557 eV results in the highest EY for all tandem modules when the albedo reaches 64% or higher (typical for bright sandstone or snow ground). To obtain the maximum EY for albedo = 25%, E_{g_PVK} in the range of 1.58–1.62 eV is strongly recommended.

It is known from literature that the PVK solar cells with lower E_{g_PVK} are characterized with lower voltage loss, higher stability, and maturer fabrication process.^[31–34] Therefore, employing PVK top cells with reduced E_{g_PVK} in bifacial tandem modules is favorable for large-scale application. In areas with strong DHI (e.g., Chengdu) compared to areas with strong DNI (e.g., Los Angeles), more oblique incident light results in lower photocurrent in the subcells. Whereas, the rear-side illumination on the bifacial tandem solar modules only contributes to the J_{sc_c-Si} enhancement, which is mainly determined by GHI and albedo, and is characterized by isotropy and less dependence on DNI. Thus, under relatively moderate albedo conditions (25–35%), regions with stronger DHI prefer narrower bandgaps to increase the absorption of PVK top cells.

The power-generation densities (PGDs) of bifacial tandem modules with narrow bandgap PVK ($E_{g_PVK} = 1.557$ eV) are compared under different albedo conditions (Figure 5d–f) for one summer day in Chengdu, Shanghai, and Los Angeles areas. The PGD is found to increase with increasing albedo, gradually saturating after albedo $\geq 35\%$. This trend can also be observed for EYs of bifacial tandem modules in different regions and with different E_{g_PVK} (see Figure S12 and S13, Supporting Information). This is because increasing the albedo only contributes to the photocurrent of the *c*-Si bottom cell, and after an increase to a certain level, with the PVK top cell becoming the current limiting one, further increasing the albedo only leads to the current mismatch thus limiting the rapid growth of EY. In addition, DNI shows a stronger influence than DHI in determining the PGD of bifacial modules. Compared to Shanghai and Los Angeles, Chengdu shows the lowest DNI during day time thus its PGD also being the lowest. Meanwhile the fluctuation of DNI from 12:00 to 16:00 in Chengdu correlates well with the fluctuation of PGD.

Since there is no current mismatch issue for bifacial SHJ modules, their EY increases linearly with the growth of albedo (see Figure S13, Supporting Information). In regions with large DHI such as Chengdu, London, and Shanghai, the EYs of SHJ modules are higher than those of tandem modules at albedo = 88%. Especially in Chengdu, the tandem PV modules are not superior in terms of EY at 35% albedo (ground condition of concrete), and the EY is significantly smaller than that of the SHJ modules when the albedo is further increased (see Figure 5a and S12a, Supporting Information). Previous studies have shown that the manufacture cost of PVK/*c*-Si tandem PV module is about $121.18 \$ m^{-2}$, which is 35% ($31.59 \$ m^{-2}$ in absolute value) higher than that of *c*-Si modules due to the additional materials and deposition process required for PVK.^[21] The manufacture cost of single-junction *c*-Si modules is about 37% of the total installed costs.^[1] Assuming the PVK/*c*-Si tandem modules holding the same fraction of manufacture cost, their total installed cost is estimated to be approximately 13% higher than of the *c*-Si counterpart. For ease of comparison, we have deployed a ratio ($LCOE_{tandem}/LCOE_{SHJ}$) as defined by the LCOE of bifacial PVK/*c*-Si tandem system divided by the LCOE of bifacial SHJ system, the ratio $LCOE_{tandem}/LCOE_{SHJ}$ less than 1 means that the tandem system brings higher revenue. The calculation details are represented in Note S3, Supporting Information. The ratio of LCOE in six typical regions (see Figure 6 for the Chengdu and Los Angeles and Figure S14, Supporting Information, for other regions) has been calculated assuming the total installed cost of PVK/*c*-Si tandem modules being higher than that of SHJ modules by 10%, 13%, and 15%, respectively.

In conjunction with the previous discussion, the E_{g_PVK} is set to be 1.58 eV for regions with a large portion of DHI (e.g., Chengdu, London, Shanghai) and 1.62 eV for regions with a large portion of DNI (e.g., Yinchuan, Lhasa, Los Angeles). As shown in Figure 6 and S14, Supporting Information, it is seen that bifacial tandem modules possess cost advantages when installed in Los Angeles with high DNI share and albedo between 9% and 35%. And, the tandem module no longer has the advantage over SHJ module when albedo exceeds 40%. More intriguingly, deploying bifacial SHJ modules in Chengdu, which has dominant DHI, is always an economical choice regardless of the albedo value. Figure 6 also demonstrates the impact of the

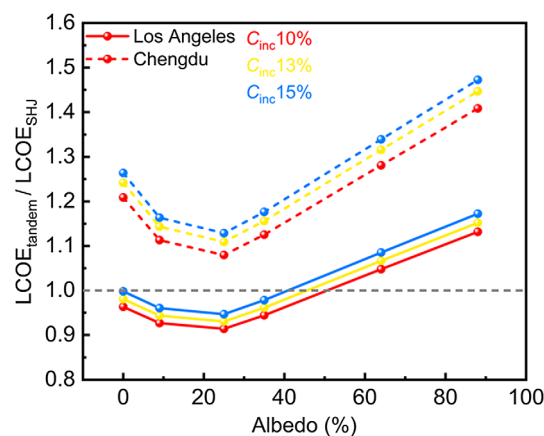


Figure 6. Ratios between levelized cost of electricity ($LCOE_{tandem}$ and $LCOE_{SHJ}$) as a function of albedo. C_{inc} represents the different installation cost increments (C_{inc} : 10% red, 13% yellow, 15% blue).

installation cost on the LCOE, that is, the higher the installation cost the greater the LCOE, which is consistent with previous reports.^[21,73]

4. Conclusion

In this work, we have investigated the EY characteristics of bifacial PVK/*c*-Si tandem modules and compared them with bifacial SHJ modules under realistic outdoor operation conditions (i.e., meteorological data of installation sites, albedo, etc.). The photocurrent of the subcells and the optimal E_{g_PVK} for monofacial and bifacial modules were obtained based on the light absorption of PVK and *c*-Si in the tandem solar cells and modules. Rear-side irradiation only boosts the photocurrent of the *c*-Si bottom cell, so increasing the rear-side irradiation intensity enables the employment of PVK materials with smaller bandgaps for more light absorption from the front side. For example, at typical ground conditions with albedo = 25%, the current matching is achieved for $E_{g_PVK} = 1.60$ eV, and the module efficiency reaches 32.49%. We then examined the EY of six typical regions under different albedo conditions. The optimal E_{g_PVK} was found to be 1.62 eV or even lower for albedo greater than 25%. In areas with a high DNI such as Yinchuan and Los Angeles, bifacial tandem modules deliver higher EY regardless of ground type with or without albedo, while in areas with a high DHI such as Chengdu, bifacial SHJ modules exhibit similar or even higher EY than that of tandem modules. Given the LCOE considerations, we recommend deploying bifacial SHJ modules under strong DHI or high albedo conditions rather than tandem ones. Overall, the bifacial tandem modules with E_{g_PVK} in the range of 1.58–1.62 eV are suitable for most application scenarios. The PVK solar cells with lower E_{g_PVK} are characterized with lower voltage loss, higher stability, and maturer fabrication process, which would further promote the large-scale application of bifacial tandem PV modules.

Bifacial PVK/*c*-Si tandem modules have the potential to generate additional energy output by receiving reflected light from the ground, thus provide an effective strategy to further improve the efficiency and EY of PV modules. However, the EY generated

by bifacial PVK/c-Si tandem modules under realistic outdoor operation varies with installation locations and conditions (such as albedo etc.). Our work provides important implications for policy making in determining the correct type of PV modules in delivering cost-effective renewable electricity.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

albedo, bandgaps, bifacial modules, energy yield, perovskite/silicon tandem solar cells

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